

# **Bass Real Analysis Solutions**

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# 1 Home

## 2 Bass Real Analysis Solutions

This website contains my solutions to selected exercises from Richard Bass's *Real Analysis for Graduate Students*.

### 2.1 About this project

The goal of this project is to provide clear and rigorous solutions to graduate-level real analysis exercises.

### 2.2 Author

Jae-Yun Lee

- B.S. in Mathematical Sciences, KAIST
- M.S. and Ph.D. Track in Economics, KAIST

For more information, see the About page.

## 3 About

## 4 Jae-Yun Lee

I am an independent researcher and musician from South Korea.

### 4.1 Education

- B.S. in Mathematical Sciences, KAIST
- M.S. and Ph.D. Track in Economics, Department of Management Engineering, KAIST

### 4.2 Research

Selected publication:

Jae-Yun Lee, Jiwoong Shin, and Jungju Yu

[“Communicating Attribute Importance under Competition”](#)

*Marketing Science*, 45(3), 534–555, 2026.

### 4.3 Music

Outside academia, I work as an independent musician under the name **Jayhat**.

My work combines hip-hop, R&B, and classical music, with a particular interest in orchestral composition and piano performance.

- Instagram: [\(jayhat3210?\)](#)

## **Part I**

# **Chapter 2**



## 5 Exercise 2.1

### 5.1 Solution

Consider  $X = \{1, 2, 3\}$  and  $\mathcal{M} = \{\emptyset, \{1\}, \{2\}, \{3\}, X\}$ . It is straightforward to verify that  $\mathcal{M}$  is a monotone class. However,  $\mathcal{M}$  is not a  $\sigma$ -algebra, since

$$\{1\} \cup \{2\} = \{1, 2\} \notin \mathcal{M}.$$

## 6 Exercise 2.2

### 6.1 Solution

Consider  $X = \{1, 2, 3, 4\}$ , and

$$\mathcal{A}_1 = \{\emptyset, \{1\}, \{2, 3, 4\}, X\}, \quad \mathcal{A}_2 = \{\emptyset, \{1, 2, 3\}, \{4\}, X\}.$$

It is straightforward to verify that  $\mathcal{A}_1$  and  $\mathcal{A}_2$  are  $\sigma$ -algebras. However,  $\mathcal{A}_1 \cup \mathcal{A}_2$  is not a  $\sigma$ -algebra, since

$$\{1\} \cup \{4\} = \{1, 4\} \notin \mathcal{A}_1 \cup \mathcal{A}_2.$$

## 7 Exercise 2.3

### 7.1 Solution

Consider  $X = \mathbb{N}$  and  $\mathcal{A}_n = \sigma(\{\{1\}, \{2\}, \dots, \{n\}\})$ .

Note that  $\mathcal{A}_n$  is a  $\sigma$ -algebra by construction. Moreover, since

$$\{\{1\}, \{2\}, \dots, \{n\}\} \subset \{\{1\}, \{2\}, \dots, \{n+1\}\},$$

we have  $\mathcal{A}_n \subset \mathcal{A}_{n+1}$ .

First, we establish a useful property of  $\mathcal{A}_n$ .

For each  $n \in \mathbb{N}$ , define

$$T_n = \{n+1, n+2, \dots\} \quad \text{and} \quad \mathcal{B}_n = \{E \subset \mathbb{N} : T_n \subset E \text{ or } E \cap T_n = \emptyset\}.$$

We claim that  $\mathcal{B}_n$  is a  $\sigma$ -algebra.

(1) First,  $\emptyset \in \mathcal{B}_n$ , since  $\emptyset \cap T_n = \emptyset$ .

(2) Second, suppose that  $E \in \mathcal{B}_n$ .

If  $T_n \subset E$ , then  $E^c \cap T_n = \emptyset$ . And if  $E \cap T_n = \emptyset$ , then  $T_n \subset E^c$ .

Thus, we have  $E^c \in \mathcal{B}_n$ .

(3) Third, suppose that  $E_k \in \mathcal{B}_n$  for  $k = 1, 2, \dots$ .

If there exists  $k$  such that  $T_n \subset E_k$ , then  $T_n \subset \bigcup_k E_k$ . Thus, we have  $\bigcup_k E_k \in \mathcal{B}_n$ .

Otherwise,  $E_k \cap T_n = \emptyset$  for all  $k$ . Hence,

$$\left( \bigcup_k E_k \right) \cap T_n = \bigcup_k (E_k \cap T_n) = \emptyset.$$

Therefore,  $\mathcal{B}_n$  is a  $\sigma$ -algebra.

Next, since  $\{1\}, \dots, \{n\} \in \mathcal{B}_n$ , we conclude that

$$\mathcal{A}_n = \sigma(\{\{1\}, \{2\}, \dots, \{n\}\}) \subset \mathcal{B}_n.$$

Consequently, for every  $E \in \mathcal{A}_n$ , we have  $E \cap T_n = \emptyset$  or  $T_n \subset E$ .

Using this property, we will show that  $\bigcup_k \mathcal{A}_k$  is not a  $\sigma$ -algebra.

Consider the set

$$\{2, 4, 6, \dots\} = \bigcup_n \{2n\}.$$

For any  $n \in \mathbb{N}$ , we have

$$\{2, 4, 6, \dots\} \cap T_n \neq \emptyset \quad \text{and} \quad T_n \not\subset \{2, 4, 6, \dots\}.$$

Hence,  $\{2, 4, 6, \dots\} \notin \mathcal{A}_n$  for any  $n \in \mathbb{N}$ , which implies that  $\{2, 4, 6, \dots\} \notin \bigcup_k \mathcal{A}_k$ .

On the other hand,  $\{2n\} \in \mathcal{A}_{2n} \subset \bigcup_k \mathcal{A}_k$  for each  $n \in \mathbb{N}$ .

Therefore,  $\bigcup_k \mathcal{A}_k$  is not closed under countable unions, and therefore is not a  $\sigma$ -algebra.

## 8 Exercise 2.4

### 8.1 Solution

(I think it is  $A_j \in \mathcal{M}_j$  in exercise, not  $A_j \in \mathcal{M}$ )

Consider  $A_n = \{2, 4, \dots, 2n\}$  and  $\mathcal{M}_n = \{\emptyset, A_n, \mathbb{N}\}$ . It is straightforward to verify that

- (1)  $\mathcal{M}_n$  is a monotone class, and
- (2)  $A_n \uparrow A = \{2, 4, 6, \dots\}$ .

However,  $A \neq A_n$  for every  $n$ , so,  $A \notin \mathcal{M}$ .

## 9 Exercise 2.5

### 9.1 Solution

We show that  $\mathcal{B}$  is a  $\sigma$ -algebra.

(1)  $\emptyset \in \mathcal{B}$

Since  $\emptyset \in \mathcal{A}$ , we have  $f^{-1}(\emptyset) \in \mathcal{B}$ . And we can say that  $f^{-1}(\emptyset) = \emptyset$ .

To prove this, suppose not. Then there exists  $x \in X$ , such that  $x \in f^{-1}(\emptyset)$ . That is,  $f(x) \in \emptyset$ . But it is a contradiction.

(2)  $X \in \mathcal{B}$

Since  $Y \in \mathcal{A}$ , we have  $f^{-1}(Y) \in \mathcal{B}$ . And we can say that  $f^{-1}(Y) = X$ .

To prove this, suppose not. Then there exists  $x \in X$ , such that  $x \notin f^{-1}(Y)$ . That is,  $f(x) \notin Y$ . But it is a contradiction.

(3) If  $B \in \mathcal{B}$ , then  $B^c \in \mathcal{B}$

Consider  $B \in \mathcal{B}$ . Then there exists  $A \in \mathcal{A}$ , such that  $f^{-1}(A) = B$ . Since

$$f^{-1}(A^c) = X \setminus f^{-1}(A) = B^c,$$

and  $A^c \in \mathcal{A}$ , we can conclude that  $B^c \in \mathcal{B}$ .

(4) If  $B_1, B_2, \dots \in \mathcal{B}$ , then  $\bigcup_i B_i \in \mathcal{B}$

Consider  $B_1, B_2, \dots \in \mathcal{B}$  and corresponding  $A_1, A_2, \dots \in \mathcal{A}$ . Since

$$\bigcup_i B_i = \bigcup_i f^{-1}(A_i) = f^{-1}\left(\bigcup_i A_i\right),$$

and  $\bigcup_i A_i \in \mathcal{A}$ , we can conclude that  $\bigcup_i B_i \in \mathcal{B}$ .

## 10 Exercise 2.6

### 10.1 Solution

Suppose, for contradiction, that  $\mathcal{A}$  is finite.

Since  $X \neq \emptyset$  and  $X \in \mathcal{A}$ , the assumed property implies that  $X$  can be written as a disjoint union of two nonempty sets in  $\mathcal{A}$ .

Choose one of them and call it  $A_1$ . Now apply the same property to  $A_1$ . Then  $A_1$  can be written as a disjoint union of two nonempty sets in  $\mathcal{A}$ .

Choose one of them and call it  $A_2$ . Since the other piece is also nonempty, we have

$$A_2 \subsetneq A_1.$$

Repeating this argument, we obtain a strictly decreasing sequence

$$X \supsetneq A_1 \supsetneq A_2 \supsetneq A_3 \supsetneq \dots$$

of sets in  $\mathcal{A}$ .

This is impossible if  $\mathcal{A}$  is finite. Hence  $\mathcal{A}$  is infinite.

By Exercise 2.8, every infinite  $\sigma$ -algebra is uncountable. Therefore,  $\mathcal{A}$  is uncountable.

# 11 Exercise 2.7

## 11.1 Solution

We show that  $\mathcal{A}$  is a  $\sigma$ -algebra.

(1)  $\emptyset \in \mathcal{A}$

Since  $\chi_{\emptyset}(x) = 0$  for every  $x \in X$ , it is a constant function. Hence,

$$\chi_{\emptyset} \in \mathcal{F}.$$

Therefore,  $\emptyset \in \mathcal{A}$ .

(2)  $X \in \mathcal{A}$

Since  $\chi_X(x) = 1$  for every  $x \in X$ , it is a constant function. Hence,

$$\chi_X \in \mathcal{F}.$$

Therefore,  $X \in \mathcal{A}$ .

(3) If  $A \in \mathcal{A}$ , then  $A^c \in \mathcal{A}$

Let  $A \in \mathcal{A}$ . Then  $\chi_A \in \mathcal{F}$ . Since

$$\chi_{A^c} = 1 - \chi_A,$$

and since  $1 \in \mathcal{F}$ , it follows that

$$\chi_{A^c} \in \mathcal{F}.$$

Therefore,  $A^c \in \mathcal{A}$ .

(4) If  $A_1, A_2, \dots \in \mathcal{A}$ , then  $\bigcup_i A_i \in \mathcal{A}$

First, if  $\chi_A, \chi_B \in \mathcal{F}$ , then

$$\chi_{A \cup B} = \chi_A + \chi_B - \chi_A \chi_B.$$

Since  $\chi_A, \chi_B \in \mathcal{F}$ , we have

$$\chi_{A \cup B} \in \mathcal{F}.$$



Thus, for every  $n \in \mathbb{N}$ ,

$$f_n := \chi_{\bigcup_{i=1}^n A_i} \in \mathcal{F}.$$

Moreover,

$$f_n(x) \rightarrow \chi_{\bigcup_i A_i}(x) \quad \text{for every } x \in X.$$

Since  $\mathcal{F}$  is closed under pointwise limits, we have

$$\chi_{\bigcup_i A_i} \in \mathcal{F}.$$

Therefore,

$$\bigcup_i A_i \in \mathcal{A}.$$

Hence,  $\mathcal{A}$  is a  $\sigma$ -algebra.

## 12 Exercise 2.8

### 12.1 Solution

Let  $X$  be a set and let  $\mathcal{A}$  be a  $\sigma$ -algebra on  $X$ . Suppose that  $\mathcal{A}$  is infinite. Then  $X$  must also be infinite, since if  $X$  were finite, then  $\mathcal{A} \subset \mathcal{P}(X)$  would also be finite.

We will show that  $\mathcal{A}$  has uncountably many elements. For this, we first construct a sequence of pairwise disjoint nonempty sets  $B_1, B_2, \dots \in \mathcal{A}$ . Then, we show that  $\mathcal{P}(\mathbb{N})$  injects into  $\mathcal{A}$ .

**Step 1:**

**Claim.** For  $E \in \mathcal{A}$ , if  $\{E \cap A : A \in \mathcal{A}\}$  is infinite, then there exists  $C \in \mathcal{A}$  such that

$$\emptyset \neq E \cap C \subsetneq E.$$

**Proof.** Suppose, to the contrary, that no such  $C$  exists. Then for every  $A \in \mathcal{A}$ , we have

$$E \cap A = \emptyset \quad \text{or} \quad E \cap A = E.$$

Hence,

$$\{E \cap A : A \in \mathcal{A}\} = \{\emptyset, E\},$$

which contradicts the assumption that this collection is infinite. ■

Now let  $A_1 = X$ . Then

$$\{A_1 \cap A : A \in \mathcal{A}\} = \mathcal{A}$$

is infinite.

Suppose that  $A_n \in \mathcal{A}$  has been chosen so that

$$\{A_n \cap A : A \in \mathcal{A}\}$$

is infinite. By the claim, there exists  $C \in \mathcal{A}$  such that

$$\emptyset \neq A_n \cap C \subsetneq A_n.$$

Then both  $A_n \cap C$  and  $A_n \cap C^c$  are nonempty members of  $\mathcal{A}$ .

At least one of the two collections

$$\{(A_n \cap C) \cap A : A \in \mathcal{A}\} \quad \text{and} \quad \{(A_n \cap C^c) \cap A : A \in \mathcal{A}\}$$

must be infinite; otherwise  $\{A_n \cap A : A \in \mathcal{A}\}$  would be finite.

Choose  $A_{n+1}$  to be one of  $A_n \cap C$  and  $A_n \cap C^c$  whose associated collection is infinite.

By induction, we obtain a strictly decreasing sequence

$$A_1 \supsetneq A_2 \supsetneq \dots$$

of nonempty sets in  $\mathcal{A}$ .

Define

$$B_n = A_n \setminus A_{n+1}.$$

Then each  $B_n$  is a nonempty member of  $\mathcal{A}$ , and the sets  $B_1, B_2, \dots$  are pairwise disjoint.

**Step 2:**

For each  $N \subset \mathbb{N}$ , define

$$E_N = \bigcup_{n \in N} B_n.$$

Since  $\mathcal{A}$  is a  $\sigma$ -algebra, we have  $E_N \in \mathcal{A}$ .

Moreover, since the sets  $B_n$  are pairwise disjoint and nonempty, if  $N \neq M$ , then

$$E_N \neq E_M.$$

Thus the map

$$N \mapsto E_N$$

is an injection from  $\mathcal{P}(\mathbb{N})$  into  $\mathcal{A}$ .

Since  $\mathcal{P}(\mathbb{N})$  is uncountable,  $\mathcal{A}$  is uncountable.

## 13 Exercise 2.9

### 13.1 Solution

(1) Let  $x \in \liminf_i A_i$ . Then there exists  $j \in \mathbb{N}$  such that

$$x \in \bigcap_{i=j}^{\infty} A_i.$$

Thus  $x \in A_i$  for every  $i \geq j$ , and hence  $x \in A_i$  for all but finitely many  $i$ .

Conversely, let  $x \in A_i$  for all but finitely many  $i$ . Then there exists  $i^*$  such that  $x \in A_i$  for every  $i \geq i^*$ . Hence,

$$x \in \bigcap_{i=i^*}^{\infty} A_i \subset \liminf_i A_i.$$

Therefore,

$$\liminf_i A_i = \{x : x \in A_i \text{ for all but finitely many } i\}.$$

Let  $x \in \limsup_i A_i$ . Then for every  $j \in \mathbb{N}$ , we have

$$x \in \bigcup_{i=j}^{\infty} A_i.$$

But if  $x \in A_i$  for only finitely many  $i$ , then there exists  $i^*$  such that  $x \notin A_i$  for every  $i \geq i^*$ . It follows that

$$x \notin \bigcup_{i=i^*}^{\infty} A_i,$$

which is a contradiction. Therefore,  $x \in A_i$  for infinitely many  $i$ .

Conversely, let  $x \in A_i$  for infinitely many  $i$ . Then for every  $j$ , we have

$$x \in \bigcup_{i=j}^{\infty} A_i.$$

Hence, we have

$$x \in \bigcap_{j=1}^{\infty} \bigcup_{i=j}^{\infty} A_i.$$

Therefore,

$$\limsup_i A_i = \{x : x \in A_i \text{ for infinitely many } i\}.$$

(2) Define

$$A_n = \left(-\frac{1}{n}, 1\right] \text{ if } n \text{ is odd, } A_n = \left(-1, \frac{1}{n}\right] \text{ if } n \text{ is even.}$$

For every  $n$ , we have

$$\bigcup_{i=n}^{\infty} A_i = (A_n \cup A_{n+1}) \cup (A_{n+2} \cup A_{n+3}) \cup \dots = (-1, 1] \cup (-1, 1] \cup \dots = (-1, 1].$$

Therefore,

$$\limsup_i A_i = \bigcap_{j=1}^{\infty} \bigcup_{i=j}^{\infty} A_i = (-1, 1].$$

If  $n$  is odd, we have

$$\begin{aligned} \bigcap_{i=n}^{\infty} A_i &= (A_n \cap A_{n+1}) \cap (A_{n+2} \cap A_{n+3}) \cap \dots \\ &= \left(-\frac{1}{n}, \frac{1}{n+1}\right] \cap \left(-\frac{1}{n+2}, \frac{1}{n+3}\right] \cap \dots = \{0\}. \end{aligned}$$

And if  $n$  is even, we have

$$\bigcap_{i=n}^{\infty} A_i = \left(-\frac{1}{n+1}, \frac{1}{n}\right] \cap \left(-\frac{1}{n+3}, \frac{1}{n+2}\right] \cap \dots = \{0\}.$$

Therefore,

$$\liminf_i A_i = \bigcup_{j=1}^{\infty} \bigcap_{i=j}^{\infty} A_i = \{0\}.$$

(3) For the first case, we have

$$\begin{aligned} \chi_{(\liminf A_i)}(x) = 1 &\iff x \in \liminf_i A_i \iff x \in A_i \text{ for all but finitely many } i \\ &\iff \chi_{A_i}(x) = 1 \text{ for all but finitely many } i \\ &\iff \text{there exists } i^* \text{ s.t. } \chi_{A_i}(x) = 1 \text{ for every } i \geq i^* \\ &\iff \liminf_i \chi_{A_i}(x) = 1. \end{aligned}$$

Since the two functions take values only in  $\{0, 1\}$ , the equivalence above implies that they are equal for every  $x$ . Therefore,

$$\chi_{(\liminf A_i)}(x) = \liminf_i \chi_{A_i}(x).$$

For the second case, we have

$$\begin{aligned}
\chi_{(\limsup A_i)}(x) = 1 &\iff x \in \limsup_i A_i \iff x \in A_i \text{ for infinitely many } i \\
&\iff \chi_{A_i}(x) = 1 \text{ for infinitely many } i \\
&\iff \sup_{i \geq i^*} \chi_{A_i}(x) = 1 \text{ for every } i^* \\
&\iff \limsup_i \chi_{A_i}(x) = 1.
\end{aligned}$$

Therefore,

$$\chi_{(\limsup A_i)}(x) = \limsup_i \chi_{A_i}(x).$$

**Part II**

**Chapter 3**

## 14 Exercise 3.1

### 14.1 Problem

Paraphrase the problem here.

### 14.2 Solution

Suppose that

$$A_n \uparrow A.$$

Then

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Hence the result follows.



## 15 Exercise 3.2

### 15.1 Problem

Paraphrase the problem here.

### 15.2 Solution

Suppose that

$$A_n \uparrow A.$$

Then

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Hence the result follows.

## 16 Exercise 3.3

### 16.1 Problem

Paraphrase the problem here.

### 16.2 Solution

Suppose that

$$A_n \uparrow A.$$

Then

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Hence the result follows.

## **Part III**

### **Chapter 4**

## 17 Exercise 4.1

### 17.1 Problem

Paraphrase the problem here.

### 17.2 Solution

Suppose that

$$A_n \uparrow A.$$

Then

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Hence the result follows.

## 18 Exercise 4.2

### 18.1 Problem

Paraphrase the problem here.

### 18.2 Solution

Suppose that

$$A_n \uparrow A.$$

Then

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Hence the result follows.

## 19 Exercise 4.3

### 19.1 Problem

Paraphrase the problem here.

### 19.2 Solution

Suppose that

$$A_n \uparrow A.$$

Then

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Hence the result follows.

## References